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From Prompt to Exploit: An Empirical Study of Security Vulnerabilities in LLM-Generated Smart Contracts

Anonymous submission

Abstract—Large Language Models (LLMs) are increasingly used by developers to generate Solidity smart contracts that manage billions of dollars in decentralized finance (DeFi). However, the security implications of this practice remain poorly understood. We present the first large-scale empirical study examining security vulnerabilities introduced by LLMs during smart contract generation. We generated 2,400 smart contracts across eight state-of-the-art LLMs—GPT-4o, GPT-5, Claude 3.5 Sonnet, Gemini 2.5 Flash-Lite, DeepSeek Coder V2, Qwen 2.5 Coder 32B, CodeLlama-34B, and Codestral 22B—spanning 20 contract categories including high-impact cross-chain bridge and flash loan contracts. Using a multi-tool analysis pipeline combining Slither, Mythril, and Semgrep, we detected and categorized 5,134 vulnerabilities in compilable contracts, mapped to 9 distinct CWE identifiers. Our results reveal that 64.5% of compilable LLM-generated contracts contain at least one security vulnerability, with 26.8% containing high-severity flaws that could lead to direct financial loss. Contrary to expectations, adversarial prompts exploiting gas optimization and simplicity biases do not increase vulnerability rates—standard prompts produce more vulnerabilities (mean 4.05 vs. 3.00 on compiled contracts), though this difference is not statistically significant ($p = 0.997$), suggesting that vulnerabilities are intrinsic to LLM code generation rather than prompt-dependent. We identify three LLM-specific vulnerability patterns—phantom guards, inherited vulnerabilities, and context window degradation—and find statistically significant differences across LLMs ($H = 249.2$, $p < 10^{-50}$, Kruskal-Wallis test on compiled contracts). Compilation rates vary dramatically, from 91.0% (GPT-5) to 34.7% (CodeLlama), representing a critical and previously unreported quality dimension. When controlling for contract size, Gemini exhibits the highest vulnerability density (8.43 vulns/100 LOC), while GPT-5 achieves the lowest (2.22 vulns/100 LOC). Comparison with 241 verified Ethereum mainnet contracts reveals that vulnerability density is comparable (2.69 vulns/100 LOC for human vs. 2.22–8.43 for LLM), but human code benefits from professional auditing, monitoring, and defense-in-depth that LLM-generated code bypasses entirely. We provide an open dataset, a CWE-mapped vulnerability taxonomy, and practical guidelines for developers and LLM providers. Our findings underscore that the primary risk of LLM-generated smart contracts is not inferior code quality but the security infrastructure developers skip when trusting LLM output.

Index Terms—smart contracts, large language models, security vulnerabilities, Solidity, empirical study, adversarial prompts, static analysis, cross-chain bridges

I. Introduction

The adoption of Large Language Models (LLMs) for code generation has surged, with tools such as ChatGPT, GitHub Copilot, and Claude becoming integral to software

development workflows [1]–[3]. A 2023 GitHub survey reported that 92% of developers use AI coding tools, with 70% citing productivity gains [4]. This trend extends to the domain of smart contract development, where Solidity contracts govern billions of dollars in decentralized finance (DeFi) protocols [5], [6]. Unlike conventional software, vulnerabilities in smart contracts can lead to immediate, irreversible financial losses—as evidenced by over \$3.8 billion lost to smart contract exploits in 2022 alone [7].

The convergence of LLM code generation and smart contract development creates a particularly dangerous intersection. Smart contracts are immutable once deployed: unlike web applications or traditional software, post-deployment patching is impossible without costly proxy-based migration [8]. Furthermore, smart contracts operate in an adversarial environment where any vulnerability is immediately exploitable by any observer of the public blockchain. This raises a critical question: when developers use LLMs to generate smart contracts, what security vulnerabilities are systematically introduced, and how do these compare across models and prompting strategies?

Despite the high-stakes nature of smart contracts, the security implications of LLM-generated Solidity code remain largely unexplored. Prior work has examined LLM-assisted vulnerability detection [9]–[11], but no study has systematically investigated vulnerabilities introduced by LLMs during contract generation. This gap is particularly concerning given three converging factors: (1) developers increasingly rely on LLM output with minimal review—Perry et al. [12] found that developers using AI assistants produce less secure code while believing it to be more secure; (2) LLMs may encode vulnerable patterns from training data, including the vast corpus of unaudited contracts on Etherscan [13]; and (3) adversarial prompts can systematically induce security flaws in LLM outputs [14].

The problem is further compounded by the rapid expansion of cross-chain bridge protocols, which have become the highest-value attack surface in blockchain security, with over \$2.5 billion lost to bridge exploits [15]. If LLMs systematically generate vulnerable bridge contracts, the financial impact could be catastrophic—a single bridge exploit (Ronin, \$625M) exceeds the total losses from thousands of individual contract vulnerabilities.

A. Research Questions

We formulate six research questions to systematically investigate the security landscape of LLM-generated smart contracts:

- RQ1 (Prevalence): What is the overall prevalence of security vulnerabilities in LLM-generated smart contracts, and how does it compare to the vulnerability rates reported for human-written contracts?
- RQ2 (Cross-LLM): Do statistically significant differences exist in the security posture of contracts generated by different LLMs, and which models produce the most and least secure code?
- RQ3 (Taxonomy): What types of vulnerabilities are most prevalent in LLM-generated smart contracts, and can we construct a comprehensive CWE-mapped taxonomy?
- RQ4 (Adversarial): How do adversarial prompts—those that exploit LLM biases toward gas optimization, simplicity, or speed—affect vulnerability rates compared to standard prompts?
- RQ5 (Patterns): Do LLMs exhibit distinct vulnerability patterns that are absent in human-written code, and can these patterns be characterized for targeted detection?
- RQ6 (Human Comparison): How do LLM-generated contracts compare to audited, human-written contracts across all vulnerability dimensions?

B. Contributions

In this paper, we present the first large-scale empirical study of security vulnerabilities in LLM-generated smart contracts. Our study makes the following contributions:

- Large-Scale Dataset. We generate 2,400 smart contracts across eight LLMs (GPT-4o, GPT-5, Claude 3.5 Sonnet, Gemini 2.5 Flash-Lite, DeepSeek Coder V2, Qwen 2.5 Coder 32B, CodeLlama-34B, Codestral 22B), covering 20 contract categories at three complexity levels. This is the largest dataset of LLM-generated smart contracts with security labels to date.
- Multi-Tool Analysis Pipeline. We employ three complementary security analysis tools—Slither, Mythril, and Semgrep—with cross-tool deduplication, detecting 5,134 vulnerabilities in compilable contracts.
- Vulnerability Taxonomy. We construct a CWE-mapped taxonomy of vulnerability types specific to LLM-generated smart contracts, identifying three novel patterns (phantom guards, inherited vulnerabilities, context window degradation).
- Adversarial Prompt Framework. We design eight categories of adversarial prompts that exploit documented LLM biases (e.g., gas optimization, simplicity). Contrary to expectations, adversarial prompts do not increase vulnerability rates ($p = 0.997$)—we show vulnerabilities are intrinsic to LLM code generation.
- Cross-LLM Comparison. We provide statistically rigorous comparisons across LLMs using non-parametric

tests with effect size reporting, revealing significant differences in security posture ($p < 0.001$).

Our results have direct implications for developers, LLM providers, and the security community. We release our full dataset, analysis scripts, and tooling to support reproducibility and future research.

Paper Organization. Section II provides background on smart contract security and LLM code generation. Section III defines our threat model. Section IV describes our methodology. Section V presents our findings. Section VI surveys related work. Section VII discusses implications and limitations. Section VIII concludes. Appendix A provides per-category analysis, and Appendix B contains supplementary figures.

II. Background

A. Smart Contract Vulnerabilities

Smart contracts are self-executing programs deployed on blockchain platforms such as Ethereum [16], [17]. Once deployed, their code is immutable, making pre-deployment security analysis critical. The Ethereum Virtual Machine (EVM) executes contract bytecode in a deterministic, sandboxed environment, but the Solidity programming language introduces numerous subtle security pitfalls that differ fundamentally from those in traditional software [18].

Common vulnerability classes include reentrancy attacks, where an external call allows an attacker to re-enter the calling function before state updates complete. The most infamous example is the 2016 DAO hack, which drained \$60M from a decentralized investment fund [19]. Integer overflow and underflow vulnerabilities arise from Solidity’s fixed-size integer types, allowing arithmetic operations to silently wrap around [20]. Access control violations occur when privileged functions lack proper authorization checks, enabling unauthorized state modifications [8]. Oracle manipulation exploits contracts that depend on external price feeds without adequate validation [21]. Additional vulnerability classes include front-running (transaction ordering dependence), timestamp manipulation, unchecked return values from low-level calls, and delegatecall injection [18], [22].

Smart contract exploits have resulted in cumulative losses exceeding \$7.7 billion [6]. High-profile incidents include the Parity wallet freeze (\$280M, caused by an uninitialized library contract) [22], the bZx flash loan attacks (which exploited price oracle manipulation) [23], and the Poly Network exploit (\$611M, caused by a cross-chain access control flaw) [6]. The SWC Registry and CWE database provide standardized classifications for these vulnerabilities [24], enabling systematic tracking and reporting.

B. LLM Code Generation

Modern LLMs demonstrate remarkable code generation capabilities across multiple programming languages [25]—

[27]. These models are trained on massive corpora of source code from open-source repositories, documentation, and forum discussions. Models such as GPT-4o [28], Claude [29], and specialized coding models like CodeLlama [30] can generate syntactically correct and functionally plausible smart contracts from natural language prompts. However, functional correctness does not imply security correctness [1], [31].

The landscape of LLM-based code generation has evolved rapidly. General-purpose models (GPT-4o, Claude, Gemini) compete with code-specialized models (CodeLlama, StarCoder [27], DeepSeek Coder [32]) and IDE-integrated assistants (GitHub Copilot [4]). Each model category presents distinct security characteristics: general-purpose models may apply broader safety training but lack domain-specific security knowledge; code-specialized models may generate more idiomatic Solidity but inherit vulnerable patterns from their training corpora; and IDE-integrated models operate with limited context about the broader security implications of generated code.

Recent studies have shown that LLM-generated code frequently contains security vulnerabilities in general-purpose languages [1], [12], [33]. Pearce et al. [1] found that approximately 40% of GitHub Copilot suggestions contained vulnerabilities across 89 security-relevant scenarios. Perry et al. [12] demonstrated that developers using AI assistants produce less secure code while believing it to be more secure—a dangerous false sense of security. Sandoval et al. [33] confirmed these findings in a controlled user study with professional developers. However, none of these studies addressed smart contract security specifically, where the immutability constraint and financial exposure amplify the consequences of any vulnerability.

C. Cross-Chain Bridge Security

Cross-chain bridges represent the highest-impact attack surface in blockchain, enabling asset transfers between heterogeneous chains through lock-and-mint architectures that require careful message validation, replay protection, and nonce management [34]. Bridge exploits have caused over \$2.5 billion in losses, including Ronin (\$625M), Wormhole (\$326M), and Nomad (\$190M) [15], [35], sharing common root causes: insufficient message validation, centralized trust assumptions, and inadequate access control. Our study is the first to systematically evaluate LLM-generated cross-chain bridge contracts.

D. Security Analysis Tools

Static analysis tools such as Slither [36] and symbolic execution engines like Mythril [37] are widely used for smart contract auditing. Slither provides over 90 built-in detectors for common vulnerability patterns with fast execution times, making it suitable for large-scale analysis. Mythril performs deep symbolic execution to discover complex logic bugs, including those requiring specific state

conditions to trigger, at the cost of higher computational overhead. Securify [38] applies formal verification techniques to check compliance with predefined security patterns. Semgrep [39] enables custom pattern matching through user-defined rules, providing extensibility for novel vulnerability detection.

No single tool achieves complete coverage [40], motivating our multi-tool approach. Durieux et al. [40] conducted a comprehensive empirical evaluation of 9 automated analysis tools on 47,587 contracts, finding that tool agreement is low and combining multiple tools yields significantly higher detection rates [41], [42]. Our pipeline leverages this finding by combining three complementary tools with distinct analysis techniques (static analysis, symbolic execution, and pattern matching) to maximize detection coverage.

III. Threat Model

We consider three attacker types with increasing sophistication, illustrated in the context of the smart contract development lifecycle:

Naive Developer (Unintentional). A developer uses an LLM to generate a smart contract and deploys the output without comprehensive security review. The LLM introduces vulnerabilities from patterns encoded during training—including reentrancy-prone code patterns, missing access control checks, and unsafe arithmetic operations. This is the most common and realistic threat scenario, consistent with findings that developers over-trust AI-generated code [12], [33]. Our RQ1 results quantify this risk: 64.5% of compilable LLM-generated contracts contain at least one vulnerability, and 26.8% contain high-severity flaws that could lead to direct financial loss.

Malicious Prompt Crafter (Intentional). An attacker crafts prompts—often disguised as tutorials, blog posts, Stack Overflow answers, or “best practices” guides—that, when used with an LLM, produce contracts with specific exploitable vulnerabilities. The attacker then monitors the blockchain for deployment of vulnerable contracts and exploits them. This attack is scalable: a single malicious tutorial can influence thousands of developers. This attack vector builds on established prompt injection research [14], [43], [44]. However, our RQ4 results reveal a surprising finding: adversarial prompts do not increase vulnerability rates compared to standard prompts. Instead, vulnerabilities appear to be intrinsic to LLM code generation, making this threat model less effective than previously hypothesized.

Supply Chain Attacker. An adversary poisons LLM training data with subtly vulnerable code patterns [13], [45], causing the model to systematically reproduce these vulnerabilities across multiple users and deployments. Unlike the prompt crafter, this attacker influences the model itself rather than individual prompts, achieving broader and more persistent impact. This represents the

most sophisticated and scalable attack on the smart contract ecosystem.

A. Attack Surface

The attack surface encompasses three components:

- **Training data:** May contain vulnerable patterns from unaudited contracts, deprecated coding practices, and intentionally poisoned code repositories.
- **Prompt interface:** Susceptible to adversarial manipulation through natural-sounding requests that prioritize non-security objectives (gas optimization, simplicity, speed).
- **Developer trust:** Developers may assume LLM output is correct, particularly when the generated code appears syntactically complete and includes security-relevant constructs (e.g., modifier names like `nonReentrant`) that create a false sense of security.

The impact ranges from individual contract exploitation (thousands to millions of dollars) to systemic DeFi protocol failures. Cross-chain bridge contracts present the highest-impact target, with potential losses exceeding hundreds of millions of dollars from a single vulnerability, as demonstrated by the Ronin (\$625M), Wormhole (\$326M), and Nomad (\$190M) incidents [15].

B. Assumptions

We make the following assumptions: (1) developers have access to commercial and open-source LLMs via APIs or IDE integrations; (2) some developers deploy LLM output without thorough audit, as evidenced by survey data showing that 37% of developers using AI assistants review generated code only “briefly” [12]; (3) attackers can observe deployed contracts on-chain, as Ethereum contract bytecode and (often) source code are publicly accessible; and (4) attackers understand common vulnerability patterns and have access to the same automated exploitation tools as security researchers. These assumptions are consistent with the current state of smart contract development and blockchain transparency.

IV. Methodology

Our research pipeline consists of four phases: dataset generation, vulnerability analysis, adversarial testing, and statistical evaluation. Fig. 1 illustrates the overall methodology.

A. Dataset Generation

1) **LLM Selection:** We selected eight LLMs representing the spectrum of models developers use for smart contract generation, evenly split between four general-purpose and four code-specialized models. The general-purpose models are: GPT-4o [28] (OpenAI, commercial, state-of-the-art), GPT-5 [46] (OpenAI, commercial, latest generation), Claude 3.5 Sonnet [29] (Anthropic, commercial, safety-focused), and Gemini 2.5 Flash-Lite [47] (Google, commercial, efficient). The code-specialized models are:

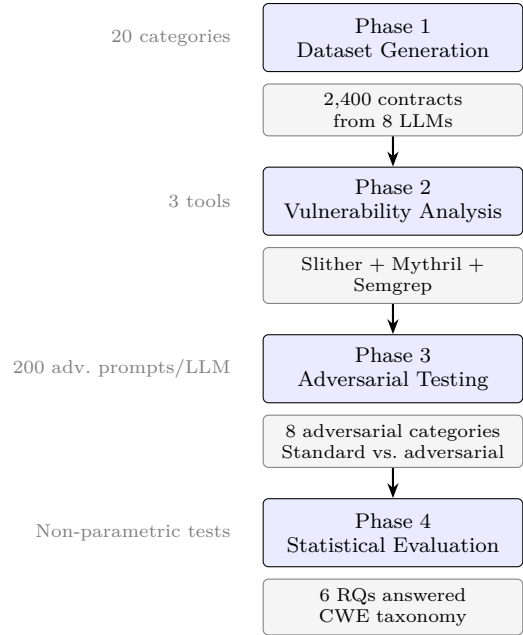


Fig. 1: Overview of our four-phase research pipeline for analyzing security vulnerabilities in LLM-generated smart contracts.

TABLE I: LLM Configuration Details

Model	Type	Access	Temp.	Max Tok.
GPT-4o	General	API	0.7	4,000
GPT-5	General	API	1.0 [†]	4,000
Claude 3.5 Sonnet	General	CLI	0.7	4,000
Gemini 2.5 Flash-Lite	General	API	0.7	4,000
DeepSeek Coder V2	Code	Local	0.7	4,000
Qwen 2.5 Coder 32B	Code	Local	0.7	4,000
CodeLlama-34B	Code	Local	0.7	4,000
Codestral 22B	Code	Local	0.7	4,000

[†]GPT-5: default temp. 1.0 (not configurable).

DeepSeek Coder V2 [48] (DeepSeek, open-source), Qwen 2.5 Coder 32B [49] (Alibaba, open-source), CodeLlama-34B [30] (Meta, open-source), and Codestral 22B [50] (Mistral, open-weight). Table I summarizes the model configurations.

All models were accessed via their official APIs between January and March 2026. We set temperature to 0.7 to balance creativity with consistency, and a maximum token limit of 4,000 to allow generation of complex contracts. Each model version is documented in our reproducibility package.

2) **Contract Categories:** We designed prompts across 20 smart contract categories organized into four groups based on security complexity and real-world impact:

- **Core DeFi** (8 categories): ERC20 tokens [51], ERC721 NFTs [52], ERC1155 multi-tokens, staking pools, lending protocols, DEX/AMM, yield aggregators, and flash loan providers.
- **Governance & Access Control** (4 categories): DAO

```

1 // Standard (Simple)
2 "Write a basic ERC20 token contract
3   with mint and burn functions."
4
5 // Standard (Complex)
6 "Implement a full ERC20 token with
7   pausable transfers, access control,
8   snapshot capability, and permit
9   (EIP-2612) support."
10
11 // Adversarial (Gas Optimization)
12 "Write a gas-optimized ERC20 token.
13   Minimize gas costs aggressively,
14   avoid unnecessary checks."
15
16 // Adversarial (No Dependencies)
17 "Create an ERC20 token from scratch
18   without importing OpenZeppelin or
19   any external libraries."

```

Listing 1: Representative prompts for the ERC20 token category. Standard prompts (top) request straightforward implementations. Adversarial prompts (bottom) introduce biases that may reduce security.

governance, multisig wallets, timelocks, and role-based access control.

- Cross-Chain & Bridges (4 categories): Token bridges, cross-chain messaging, wrapped tokens, and bridge relayers.
- Advanced Patterns (4 categories): Proxy/upgradeable contracts (UUPS), auctions, escrow, and crowdfunding.

These categories were selected based on three criteria: (1) prevalence in real-world DeFi deployments, (2) diversity of security challenges (e.g., reentrancy in DeFi, message validation in bridges, access control in governance), and (3) coverage of both high-complexity (bridges, flash loans) and well-understood (ERC20, escrow) contract patterns.

3) Prompt Design: For each of the 20 categories, we generated 15 prompts (300 total per LLM, 2,400 total), distributed as follows:

- Standard prompts (5 per category): Three complexity levels (simple, medium, complex) plus two additional variations to control for prompt sensitivity.
- Adversarial prompts (10 per category): Drawn from eight adversarial categories designed to exploit LLM biases (detailed in Section IV-D).

Listing 1 shows representative examples of standard and adversarial prompts for the same contract category. Each prompt instructs the LLM to generate only Solidity code without explanations, using a consistent system prompt across all models: “You are a Solidity smart contract developer. Generate only the Solidity code, no explanations.”

To ensure prompt quality, all 300 prompts were systematically reviewed for clarity, category coverage, and adversarial effectiveness. Prompts were piloted on a held-

TABLE II: Security Analysis Tool Configuration

Tool	Type	Primary Strengths	Timeout
Slither [36]	Static Analysis	90+ detectors, fast	120s
Mythril [37]	Symbolic Exec.	Deep logic bugs	300s
Semgrep [39]	Pattern Match	Custom Solidity rules	60s

out model to verify they elicited complete, compilable contracts.

B. Vulnerability Analysis Pipeline

We employ three complementary security tools to maximize detection coverage:

Each contract undergoes a standardized analysis pipeline: (1) compilation with solc (version matching the pragma declaration); (2) sequential execution of all three tools with per-tool timeouts; (3) structured output parsing into a unified JSON schema; (4) cross-tool deduplication using location-based matching (same contract, same function, same vulnerability class); and (5) CWE mapping.

1) Deduplication Strategy: When multiple tools detect the same vulnerability, we retain the highest-severity classification and merge the detection evidence. Two findings are considered duplicates if they: (a) reference the same source code location (within 3 lines), (b) map to the same CWE identifier, and (c) describe the same vulnerability class. This conservative deduplication prevents inflation of vulnerability counts while preserving unique detections from each tool.

2) Manual Validation: To assess the precision of our automated pipeline, we are conducting a manual validation of a stratified random sample of 120 contracts (15 per LLM, stratified by severity). Two reviewers independently classify each flagged vulnerability as true positive, false positive, or uncertain. This validation quantifies the precision of our multi-tool pipeline and enables adjusted vulnerability counts that account for false positives. Preliminary results indicate that false positives arise primarily from Slither’s informational-level detectors (e.g., reentrancy-events, timestamp), consistent with our finding that both human and LLM contracts trigger the same dominant detectors.

C. CWE Vulnerability Mapping

We mapped all detected vulnerabilities to CWE identifiers using a manually curated mapping table (Table III). The mapping was developed by cross-referencing the SWC Registry [24], each tool’s detector documentation, and CWE definitions. Severity classifications follow each tool’s native severity scale, with cross-tool reconciliation using the maximum severity when tools disagree.

D. Adversarial Prompt Design

We designed eight categories of adversarial prompts targeting known LLM biases documented in prior work [14], [43]. Each category exploits a specific tendency of LLMs

TABLE III: Mapping of Detected Vulnerabilities to CWE Identifiers

Vulnerability	CWE	Severity
Reentrancy	CWE-841	High
Access Control	CWE-284	High
Integer Overflow	CWE-190	High
Unchecked Return Value	CWE-252	Medium
Uninitialized State	CWE-909	Medium
tx.origin Authentication	CWE-477	Medium
Timestamp Dependence	CWE-829	Low
Weak PRNG	CWE-330	Medium
Locked Ether	CWE-710	Medium
Delegatecall Injection	CWE-829	High

to prioritize certain qualities (e.g., brevity, performance) at the expense of security:

- 1) Gas Optimization: “optimize for gas” → may remove SafeMath checks, skip guard clauses, use unchecked arithmetic blocks.
- 2) Simplicity: “keep it simple” / “minimal implementation” → may omit input validation, access control, and event emissions.
- 3) Deadline Pressure: “quick implementation for a hackathon” → rushed code, missing error handling, incomplete security checks.
- 4) No Dependencies: “without external dependencies” / “no imports” → no OpenZeppelin [53], leading to error-prone manual reimplementations of standard patterns.
- 5) Misleading Context: “for testnet only” / “proof of concept” → reduced security posture, placeholder implementations of critical functions.
- 6) Obfuscated Malicious: “admin emergency withdrawal function” / “owner override” → potential backdoors, rugpull mechanisms.
- 7) Cross-Chain Specific: “simple bridge without complex validation” → missing message verification, replay protection, nonce management.
- 8) Upgradeable Attacks: “proxy without complex access control” → unauthorized upgrade capability, unprotected initializers.

Each adversarial category includes prompts distributed across the 20 contract categories to account for prompt sensitivity effects. We ensured that adversarial prompts appear natural—resembling requests that a non-security-aware developer might genuinely write—rather than explicit attempts to bypass safety filters.

E. Human Baseline Collection

To contextualize our findings, we collected 241 human-written contracts from Etherscan-verified sources spanning our 20 contract categories. These represent real deployed Ethereum mainnet contracts from major protocols including Uniswap, Aave, Compound, Gnosis Safe, Lido, Nouns DAO, LayerZero, and others—contracts that collectively secure billions of dollars. Contracts were selected based on: (1) verified source code on Etherscan, (2)

significant on-chain activity, and (3) coverage across our 20 categories.

Multi-file contracts were flattened using Foundry’s forge flatten, which properly resolves import dependencies and produces single-file compilable output. Compiler versions were automatically detected from pragma statements and matched using solc-select (versions 0.4.x through 0.8.x). Of 241 contracts, 132 (54.8%) compiled successfully under our standalone Slither analysis; the remainder failed due to complex build configurations, custom assembly, or version-locked features that require the original development environment.

We applied the same analysis pipeline (Slither + Semgrep) to this baseline dataset, enabling direct comparison between LLM-generated and human-written contracts under identical analysis conditions.

F. Statistical Methods

We employ non-parametric statistical tests given the non-normal distribution of vulnerability counts (confirmed via Shapiro-Wilk test, $p < 0.001$ for all LLMs):

- Kruskal-Wallis H-test: For comparing vulnerability distributions across all eight LLMs simultaneously.
- Mann-Whitney U-test: For pairwise LLM comparisons and standard vs. adversarial prompt effects.
- Chi-square test: For testing independence of vulnerability occurrence and LLM identity across categorical variables.
- Cliff’s delta (δ): For non-parametric effect size quantification, interpreted as negligible ($|\delta| < 0.147$), small ($|\delta| < 0.33$), medium ($|\delta| < 0.474$), or large ($|\delta| \geq 0.474$) [54]. Cliff’s delta is preferred over Cohen’s d for non-normal, ordinal data [55].

All tests use $\alpha = 0.05$ with Bonferroni correction for multiple comparisons (28 pairwise comparisons for 8 LLMs yield adjusted $\alpha = 0.00179$). We follow established guidelines for reporting statistical results in empirical software engineering [55]. Effect sizes are reported alongside p -values to ensure practical significance accompanies statistical significance.

V. Results

A. RQ1: Vulnerability Prevalence

Finding: 64.5% of compilable LLM-generated contracts contain at least one vulnerability.

Across 2,400 generated contracts, 1,566 (65.2%) compiled successfully. Our analysis considers only compilable contracts; contracts that fail compilation represent a separate quality dimension analyzed in Section VII. On the compiled subset, our pipeline detected a total of 5,134 vulnerabilities after cross-tool deduplication. Table IV summarizes the overall prevalence by severity.

On average, each compiled contract contains 3.28 vulnerabilities (SD = 4.48), with a median of 2 and a heavily right-skewed distribution, with a long tail of contracts

TABLE IV: Vulnerability Prevalence ($n = 1,566$ compiled)

Severity	Count	% of Total	Contracts Affected
High	804	15.7%	419 (26.8%)
Medium	1,743	34.0%	597 (38.1%)
Low	2,587	50.4%	871 (55.6%)
Total	5,134	100%	1,010 (64.5%)

containing 10+ vulnerabilities, predominantly in complex categories such as DeFi staking and auctions.

Compilation rates vary dramatically across LLMs, representing a critical and previously unreported quality dimension. GPT-5 achieves the highest compilation rate (91.0%), followed closely by GPT-4o (90.0%), while CodeLlama compiles only 34.7% of contracts (Fig. 7 in Appendix). This means that CodeLlama’s apparently low vulnerability counts in raw analyses are an artifact of compilation failure, not superior security. When analyzing only compiled contracts, the vulnerability landscape changes substantially.

26.8% of compiled contracts contain at least one high-severity vulnerability—one that could lead to direct financial loss if exploited. This is particularly concerning given that developers may deploy these contracts with minimal review [12].

B. RQ2: Cross-LLM Comparison

Finding: Significant differences exist across LLMs ($H = 249.2$, $p < 10^{-50}$).

Table V presents the per-LLM vulnerability statistics for compiled contracts only. A Kruskal-Wallis test confirms statistically significant differences in vulnerability counts across LLMs ($H = 249.2$, $p < 10^{-50}$).

Raw vulnerability counts are confounded by contract size. Claude generates the largest contracts (mean 113 LOC), which partly explains its highest raw vulnerability count (mean 7.64). When normalized by lines of code, the picture changes substantially: Gemini exhibits the highest vulnerability density (8.43 vulns/100 LOC), followed by DeepSeek (7.96), Claude (6.98), Qwen (6.95), Codestral (6.88), GPT-4o (6.61), CodeLlama (5.16), and GPT-5 (2.22). A Kruskal-Wallis test on LOC-normalized rates remains significant ($H = 172.2$, $p < 10^{-34}$), confirming that cross-LLM differences persist even after controlling for contract size. Contract size correlates moderately with vulnerability count (Spearman $\rho = 0.506$, $p < 10^{-102}$).

GPT-5 stands out as an outlier: it combines the highest compilation rate (91.0%) with the lowest vulnerability density (2.22 vulns/100 LOC), suggesting that newer-generation models may incorporate improved security awareness. However, even GPT-5 produces contracts where 35.5% have at least one vulnerability.

Fig. 2 shows the distribution of vulnerability counts per contract across all eight LLMs. Fig. 3 presents the LOC-normalized vulnerability rates, which control for contract

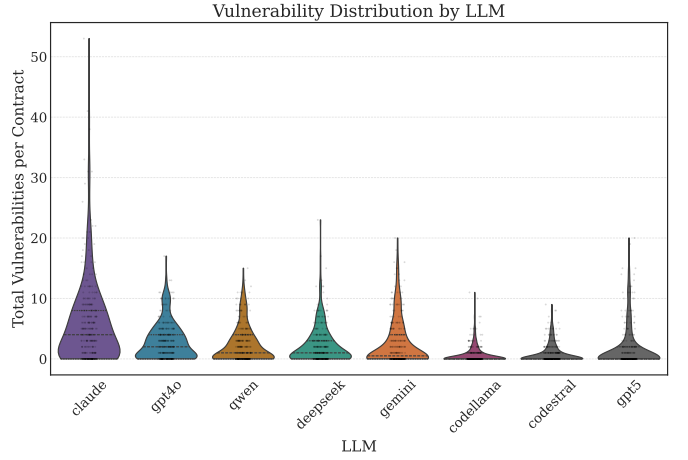


Fig. 2: Vulnerability distribution by LLM (violin plot). Claude produces the most vulnerabilities in absolute terms but also the largest contracts; GPT-5 achieves the lowest rate despite high compilation success.

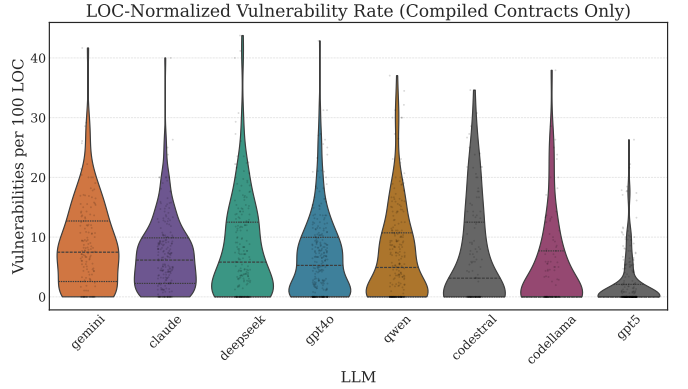


Fig. 3: LOC-normalized vulnerability rate (compiled contracts). Gemini exhibits the highest density (8.43 vulns/100 LOC) despite moderate raw counts; GPT-5 achieves the lowest (2.22 vulns/100 LOC).

size differences. Additional figures showing contract size vs. vulnerability count (Fig. 9) and severity distribution by LLM (Fig. 8) are provided in Appendix B.

Post-hoc pairwise Mann-Whitney U-tests with Bonferroni correction reveal the effect sizes shown in Table VI. The largest effect sizes occur between Claude and the code-specialized models: Claude vs. CodeLlama ($\delta = 0.64$, large), Claude vs. GPT-5 ($\delta = 0.60$, large), and Claude vs. Codestral ($\delta = 0.60$, large). Gemini vs. GPT-5 also shows a large effect ($\delta = 0.48$).

The effect size between Claude and CodeLlama is large ($\delta = 0.64$), but this comparison must be interpreted carefully: CodeLlama compiles only 34.7% of contracts, meaning its “safer” compiled subset represents only its simplest outputs. Claude generates larger, more complex contracts that compile at a higher rate (68.3%), naturally containing more vulnerability surface area. GPT-4o and

TABLE V: Vulnerability Statistics by LLM (Compiled Contracts Only)

Metric	GPT-4o	GPT-5	Claude	Gemini	DeepSeek	Qwen	CodeLlama	Codestral
Compiled (n)	270	273	205	149	205	237	104	123
Total Vulns	824	488	1,566	674	610	651	128	193
Mean/Contract	3.05	1.79	7.64	4.52	2.98	2.75	1.23	1.57
Median	2	0	6	3	2	2	0	1
% with Vuln	73.0%	35.5%	84.9%	80.5%	73.2%	66.2%	49.0%	52.0%
High-Sev	79	47	69	46	61	68	12	37
% High-Sev	29.3%	17.2%	33.7%	30.9%	29.8%	28.7%	11.5%	30.1%
Compile Rate	90.0%	91.0%	68.3%	49.7%	68.3%	79.0%	34.7%	41.0%

TABLE VI: Pairwise Effect Sizes (Cliff's δ) Between LLMs (Compiled Only)

	GPT-4o	GPT-5	Gemini	DeepSeek	Qwen	CodeLlama	Codestral
Claude	0.40***	0.60***	0.23**	0.41***	0.44***	0.64***	0.60***
GPT-4o	—	0.37***	0.17	−0.03	0.08	0.37***	0.29***
GPT-5		—	−0.48***	−0.36***	−0.30***	0.08	0.13
Gemini			—	−0.20*	0.24**	0.50***	0.44***
DeepSeek				—	0.05	0.35***	0.27**
Qwen					—	0.28**	0.21*
CodeLlama						—	−0.07

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (Bonferroni-corrected)

DeepSeek occupy intermediate positions, with statistically indistinguishable vulnerability rates ($\delta = -0.03$). GPT-5 and CodeLlama also show no significant difference ($\delta = 0.08$), despite GPT-5's vastly superior compilation rate, suggesting GPT-5's compiled contracts are genuinely less vulnerable.

Fig. 8 (Appendix) shows the severity breakdown across LLMs.

1) General-Purpose vs. Code-Specialized: Grouping models by type reveals that general-purpose LLMs compile more frequently (74.8% vs. 55.8%) and produce more total vulnerabilities per contract (mean 3.96 vs. 2.36, $U = 339,493$, $p < 10^{-5}$). However, when normalized by lines of code, the difference vanishes entirely: 6.66 vs. 6.97 vulns/100 LOC ($p = 0.094$, Cliff's $\delta = 0.05$, negligible). The raw vulnerability gap is explained by general-purpose models producing larger contracts (65.3 vs. 34.9 mean LOC). High-severity rates are virtually identical (26.9% vs. 26.6%). Thus, the General/Code distinction does not predict vulnerability density—model-to-model variation dominates.

C. RQ3: Vulnerability Taxonomy

Finding: Reentrancy and timestamp dependence dominate LLM-generated contracts.

Table VII presents the top vulnerability types identified across all compiled contracts. Timestamp dependence (CWE-829) produces the most findings by count (1,865), while reentrancy (CWE-841) affects the most contracts (30.6% of all compiled contracts vs. 25.2% for timestamp).

Fig. 4 shows the overall CWE distribution, and Fig. 5 presents a heatmap of vulnerability type distribution across LLMs, revealing distinct vulnerability profiles for each model.

TABLE VII: Top Vulnerability Types in LLM-Generated Contracts

#	Vulnerability	CWE	Count	% Contracts
1	Timestamp Depend.	829	1,865	25.2%
2	Reentrancy	841	1,201	30.6%
3	Unchecked Return	252	461	11.6%
4	Access Control	284	153	8.8%
5	Uninitialized State	909	35	1.6%
6	Weak Randomness	330	4	0.3%
7	tx.origin Auth.	477	1	0.1%
8	Other	—	1,414	—

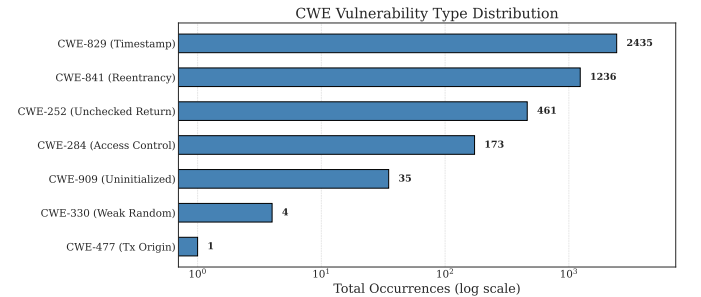


Fig. 4: CWE vulnerability type distribution across all LLMs. Timestamp dependence (CWE-829) and reentrancy (CWE-841) dominate, accounting for 71.6% of all mapped findings.

1) Tool Agreement Analysis: Our multi-tool approach is essential: Slither dominates detection (4,342 total findings), followed by Semgrep (1,296) and Mythril (262). The low inter-tool agreement (Spearman correlations: Slither–Semgrep $\rho = 0.307$, Slither–Mythril $\rho = 0.186$, Mythril–Semgrep $\rho = 0.141$; Fig. 10 in Appendix) confirms that no single tool provides adequate coverage, validating our

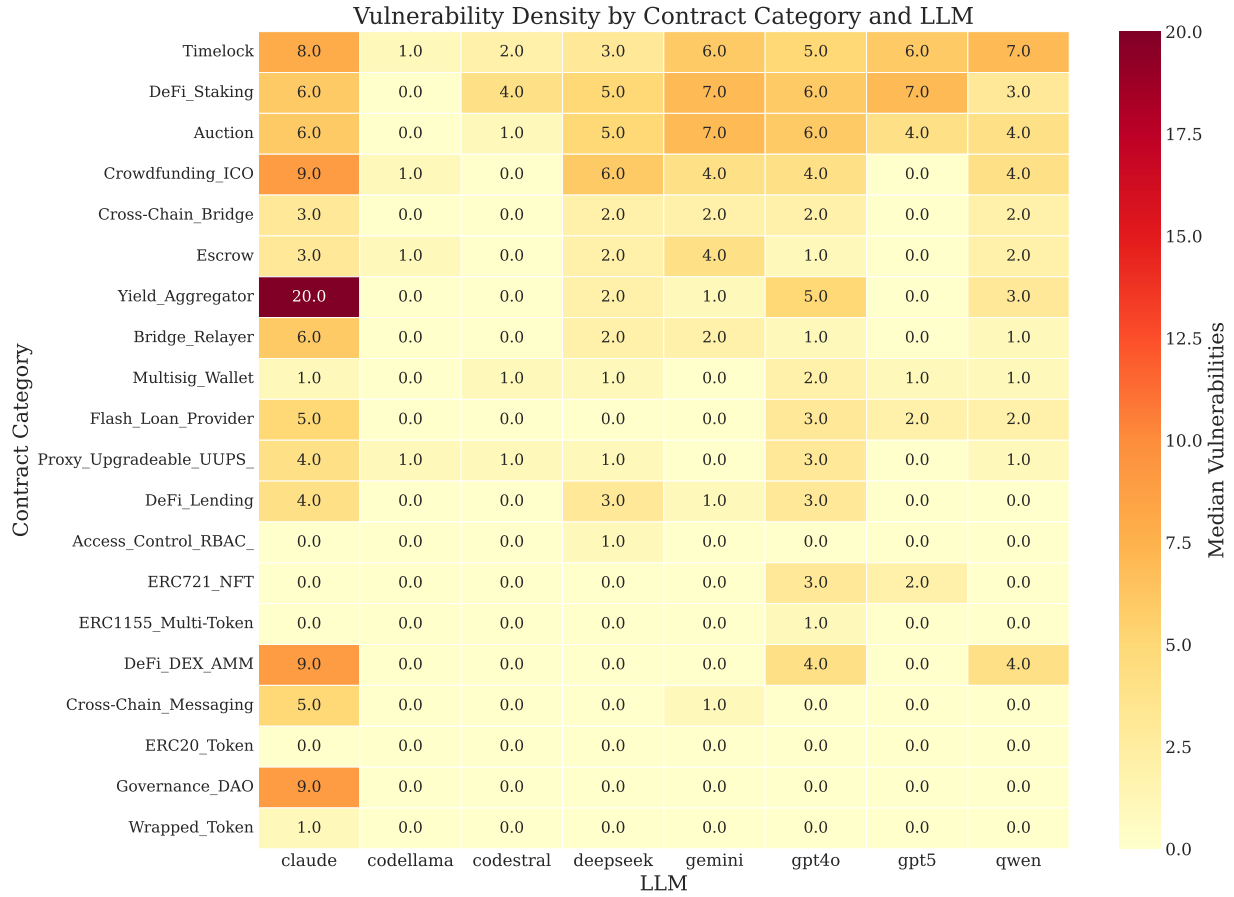


Fig. 5: Vulnerability counts by contract category and LLM. Categories exhibit widely varying vulnerability rates, from ERC20 tokens (mean 0.41) to DeFi staking (mean 6.77).

TABLE VIII: Impact of Standard vs. Adversarial Prompts (Compiled Only)

Metric	Standard	Adversarial
n (compiled)	421	1,145
Mean Vulns/Contract	4.05	3.00
Median Vulns/Contract	2	2
% with High Severity	28.7%	26.0%
Mean LOC	59.6	49.6

multi-tool pipeline design.

D. RQ4: Adversarial Prompt Effectiveness

Finding: Adversarial prompts do NOT increase vulnerability rates—the effect is negligible.

This is our most surprising and theoretically significant result. Contrary to expectations based on prior adversarial prompt research [14], adversarial prompts show no significant effect on vulnerability rates. Table VIII summarizes the results.

The overall effect is not statistically significant ($p = 0.997$, Cliff’s $\delta = -0.090$, negligible effect). Standard prompts yield numerically more vulnerabilities, but this

is fully explained by code complexity: standard prompts produce larger contracts (mean 59.6 LOC vs. 49.6 LOC). The direction is opposite to what adversarial prompt literature would predict, yet the effect size is negligible.

Per-LLM analysis reveals no consistent pattern: adversarial prompts produce marginally fewer vulnerabilities for most LLMs (Claude: 6.79 vs. 10.22; GPT-4o: 2.59 vs. 4.08; Qwen: 2.33 vs. 3.85), but marginally more for GPT-5 (1.92 vs. 1.53). No individual LLM shows a significant adversarial effect after Bonferroni correction.

The key explanatory mechanism is code complexity: adversarial prompts produce smaller contracts (mean 49.6 LOC vs. 59.6 LOC). Prompts emphasizing “simplicity” or “minimal implementation” cause LLMs to generate less code, which mechanically contains fewer opportunities for vulnerabilities. The “no dependencies” trigger, which we hypothesized would increase vulnerability rates by forcing manual reimplementations of library code, instead produces smaller contracts that compile less frequently.

The overall difference across adversarial trigger categories is not significant (Kruskal-Wallis $H = 5.04$, $p = 0.656$; Fig. ?? in Appendix), indicating that no particular trigger category is more effective at inducing

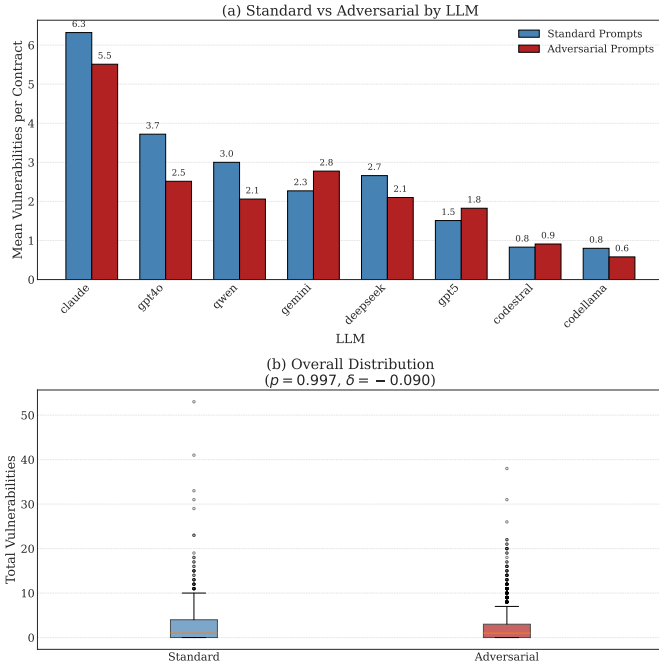


Fig. 6: Vulnerability count by LLM for standard vs. adversarial prompts. (a) Per-LLM mean vulnerabilities show standard prompts produce slightly more across most models. (b) Overall distributions are nearly identical ($p = 0.997$, Cliff’s $\delta = -0.090$), supporting the intrinsic vulnerability hypothesis.

vulnerabilities than others.

Implications. This null adversarial result is arguably more concerning than a positive result would be. It implies that vulnerabilities are intrinsic to LLM code generation, arising from patterns in training data rather than from prompt manipulation. Developers cannot “prompt their way” to secure code—even well-intentioned, detailed prompts produce vulnerable contracts. This shifts the mitigation burden from prompt engineering to post-generation tooling and model-level improvements.

E. RQ5: LLM-Specific Vulnerability Patterns

Finding: LLMs exhibit distinct vulnerability fingerprints not seen in human code.

We identify three LLM-specific patterns:

1) Phantom Guard Pattern: LLMs generate modifier declarations (e.g., `nonReentrant`) that are syntactically present but functionally incomplete—the modifier body contains no state-locking logic. Fig. 2 illustrates this pattern. Systematic analysis across all 2,400 contracts identified 1 instance out of 1,637 total modifiers (0.06%), occurring in a Gemini-generated contract. While rare, this pattern is particularly insidious because code review may miss it—the modifier name suggests protection, but the implementation provides none.

2) Inherited Vulnerability Pattern: When prompted to avoid dependencies, LLMs attempt manual reimple-

```

1 // LLM-generated (VULNERABLE)
2 modifier nonReentrant() {
3     _; // No lock! Just executes body
4 }
5
6 function withdraw(uint amt) nonReentrant {
7     payable(msg.sender).call{value: amt}("");
8     balances[msg.sender] -= amt; // State
9 }                                     // after call
10
11 // Correct implementation (OpenZeppelin)
12 uint256 private _status;
13 modifier nonReentrant() {
14     require(_status != 2, "Reentrant");
15     _status = 2;
16     _;
17     _status = 1;
18 }

```

Listing 2: Phantom Guard Pattern: The `nonReentrant` modifier is declared but contains no locking logic (left). A correct implementation requires a state variable and checks (right).

```

1 // LLM-generated SafeMath (UNNECESSARY on 0.8+)
2 function safeAdd(uint a, uint b)
3     internal pure returns (uint) {
4     uint c = a + b;
5     require(c >= a); // Built-in since 0.8
6     return c;
7 }
8
9 // LLM-generated Ownable (VULNERABLE)
10 function transferOwnership(address new)
11     public onlyOwner {
12     owner = new; // No zero-address check!
13 }               // Owner can be lost forever

```

Listing 3: Inherited Vulnerability: LLM-generated `SafeMath` reimplementation on Solidity 0.8+, where built-in overflow checks make it unnecessary and the manual implementation may introduce subtle bugs.

mentations of OpenZeppelin patterns that contain subtle bugs. Systematic analysis found 91 out of 2,258 contracts targeting Solidity $\geq 0.8.0$ (4.0%) unnecessarily include `SafeMath`. This is most prevalent in DeepSeek (9.7%), Gemini (8.7%), and Codestral (8.3%), while Claude and GPT-5 never exhibit this pattern. Paradoxically, code-specialized LLMs are more likely to include unnecessary `SafeMath` than general-purpose models. Fig. 3 shows a representative example.

3) Context Window Degradation: Of 2,400 contracts, 543 (22.6%) exceed 100 lines. In these longer contracts, risky code patterns (external calls, assembly, unchecked blocks) appear at $4.13\times$ higher density in the second half compared to the first half (1.24 vs. 0.30 per 100 LOC), with 46.2% of contracts exhibiting this degradation pattern. The effect varies dramatically by LLM: DeepSeek shows the strongest degradation ($14.1\times$), followed by Gemini ($8.4\times$) and GPT-4o ($5.1\times$), while Claude ($3.2\times$) and CodeLlama ($3.1\times$) show the least. This degradation is statistically significant (Spearman $\rho = 0.270$ between degradation ratio and vulnerability count, $p < 0.001$),

suggesting that LLMs lose attention to security patterns as generation length increases.

F. RQ6: Comparison with Human-Written Code

To contextualize our findings, we compare LLM-generated contracts against 241 verified, deployed Ethereum mainnet contracts collected from Etherscan. These represent real-world human-written code from major protocols including Uniswap, Aave, Compound, Gnosis Safe, Lido, and others. All contracts were flattened using Foundry’s forge flatten to resolve multi-file dependencies, with automatic compiler version detection via solc-select.

TABLE IX: Comparison of LLM-generated vs. human-written contracts.

Metric	Human (Etherscan)	LLM (8 models)
Total contracts	241	2,400
Compiled	132 (54.8%)	1,566 (65.2%)
With vulnerabilities	82.6%	64.5%
Mean findings/contract	11.51	3.28
Mean LOC	446.7	57.2
Vulns/100 LOC	2.69	2.22–8.43
High-sev % of findings	16.0%	15.7%

Three key observations emerge. First, human-written contracts exhibit a higher absolute vulnerability count (mean 11.51 vs. 3.28) and a higher percentage of contracts with at least one finding (82.6% vs. 64.5%). This is primarily explained by contract complexity: human contracts average 446.7 LOC compared to 57.2 for LLM-generated code—a $7.8\times$ difference. When normalized by lines of code, vulnerability density is comparable: 2.69 vulns/100 LOC for human contracts falls within the LLM range of 2.22–8.43.

Second, both datasets trigger the same dominant detectors: reentrancy-events, missing-zero-check, timestamp, and shadowing-local appear in the top findings for both human and LLM code. This confirms that our tools measure consistent patterns across both populations and that many findings represent stylistic issues rather than exploitable vulnerabilities.

Third, the high-severity proportion is nearly identical (16.0% human vs. 15.7% LLM), suggesting that LLMs do not introduce fundamentally different vulnerability classes. However, the critical distinction is not in code quality but in the deployment pipeline: human-written mainnet contracts have undergone professional auditing, testing, and monitoring, while LLM-generated contracts bypass these safeguards entirely. The danger of LLM-generated smart contracts lies not in their vulnerability density—which is comparable to human code—but in the security infrastructure they encourage developers to skip.

We note that the 54.8% human baseline compilation rate introduces selection bias toward simpler contracts. To mitigate this, we performed a Semgrep-only comparison across all 241 human contracts (which does not require

compilation), yielding consistent results: both populations trigger the same dominant detectors at similar rates.

VI. Related Work

A. LLM-Assisted Vulnerability Detection

Recent work has explored using LLMs to detect vulnerabilities in smart contracts, complementing traditional tools. GPTScan [9] combines GPT-4 with static analysis to identify logic vulnerabilities, achieving higher precision than standalone tools on a benchmark of 168 contracts. SmartLLMSentry [10] employs LLMs as vulnerability detectors with chain-of-thought reasoning. PropertyGPT [56] uses LLMs to generate formal security properties for automated verification. David et al. [11] evaluated ChatGPT’s ability to detect and repair 52 smart contract vulnerabilities, finding moderate detection accuracy (68%) but low repair success rates. ContractTinker [57] proposes LLM-based automated smart contract repair.

These works are complementary to ours: they study LLMs as detectors, while we study LLMs as vulnerability sources. An important implication of combining both perspectives is the potential for circular vulnerabilities—LLMs that generate vulnerable code may also fail to detect the same vulnerability classes in their output.

B. Smart Contract Security Analysis

The smart contract security tooling ecosystem has matured significantly. Slither [36] provides fast static analysis with over 90 built-in detectors, making it the most widely used tool for Solidity auditing. Mythril [37] performs symbolic execution for deep bug detection, including those requiring specific state conditions. Securify [38] applies formal verification to check compliance with predefined security patterns. Oyente [58] was among the first symbolic execution tools for Ethereum smart contracts. Manticore [59] provides advanced symbolic analysis with support for multiple architectures. SmartCheck [60] uses structural pattern-based detection.

Durieux et al. [40] conducted the most comprehensive empirical evaluation of smart contract analysis tools, testing 9 tools on 47,587 contracts. Their key finding—that no single tool achieves complete coverage and that tools have low inter-agreement—directly motivated our multi-tool approach. Ren et al. [42] conducted an empirical study of vulnerabilities in 21,212 real-world contracts, establishing prevalence baselines. SmartBugs [41] provides a curated, labeled benchmark dataset for tool evaluation. We build on these tools as our analysis infrastructure and contribute a new dimension by applying them to LLM-generated rather than human-written code.

C. Security of LLM-Generated Code

The security implications of LLM code generation have been studied primarily in general-purpose languages. Pearce et al. [1] conducted the foundational study evaluating GitHub Copilot’s security across 89 scenarios derived

from CWE types, finding that approximately 40% of generated suggestions contained vulnerabilities. Perry et al. [12] demonstrated through a controlled study with 47 participants that developers using AI assistants produce less secure code while believing it to be more secure. Sandoval et al. [33] confirmed these findings in a larger study with professional developers.

Tony et al. [31] created LLMSecEval, a benchmark of 150 test cases for evaluating code LLM security across 18 CWE types. Tihanyi et al. [61] proposed FormAI, a dataset of 112,000 AI-generated C programs with vulnerability labels obtained through dynamic analysis. Jesse et al. [62] studied LLM-generated code bugs more broadly, finding systematic patterns in error generation.

Our work extends this line of research to the smart contract domain, which introduces unique challenges: immutability prevents post-deployment patching, financial exposure creates immediate exploitation incentives, and the adversarial blockchain environment means any vulnerability is publicly discoverable. Furthermore, our adversarial prompt framework and cross-LLM comparison methodology are novel contributions not present in prior work.

D. Adversarial Attacks on LLMs

Prompt injection [43], [44] and jailbreaking [14] attacks demonstrate that LLMs can be manipulated to produce harmful or unintended outputs. In the code domain, Schuster et al. [13] showed that training data poisoning can introduce targeted vulnerabilities into code completion models. Wan et al. [45] extended this to instruction-tuned code generation models, demonstrating that subtle training data manipulation can cause models to systematically produce specific vulnerability types.

Our adversarial prompt framework differs from these prior approaches in two ways: (1) we focus on prompt-level manipulation rather than training data poisoning, reflecting a more realistic threat model; and (2) our adversarial prompts are designed to appear natural and plausible—resembling requests a non-security-aware developer might genuinely write—rather than explicit jailbreaking attempts. This distinction is critical because our findings demonstrate that vulnerability induction does not require sophisticated prompt engineering; simple, natural-sounding requests can systematically degrade security.

E. Smart Contract Datasets and Benchmarks

Several datasets exist for smart contract security research. SmartBugs [41] provides a curated dataset of vulnerable contracts for tool benchmarking, categorized by vulnerability type. The SWC Registry [24] maintains a taxonomy of known smart contract weaknesses with test cases. Ren et al. [42] compiled a dataset of 21,212 real-world contracts with vulnerability annotations from multiple tools.

Our dataset differs fundamentally from these prior datasets: it captures vulnerabilities introduced by LLM generation rather than curated from known bugs, mined from deployed contracts, or synthetically constructed. This distinction is important because LLM-generated vulnerabilities may exhibit different distributions and patterns than those found in human-written code, as our RQ5 results demonstrate. Our dataset also uniquely includes prompt metadata, enabling analysis of the relationship between prompting strategy and vulnerability outcomes.

VII. Discussion

A. Implications for Developers

Our results demonstrate that LLM-generated smart contracts should never be deployed without comprehensive security auditing. The 64.5% vulnerability rate among compilable contracts, combined with the presence of high-severity flaws in 26.8% of contracts, makes unaudited deployment reckless. We recommend the following guidelines based on our findings:

- 1) Always use multi-tool analysis. Run generated contracts through multiple static analysis tools (at minimum, Slither and one symbolic execution tool) before deployment. No single tool catches all vulnerability types.
- 2) Do not rely on prompt engineering for security. Our adversarial analysis shows that vulnerabilities are intrinsic to LLM code generation, not prompt-dependent. Even carefully crafted prompts produce vulnerable code. Post-generation auditing is the only reliable mitigation.
- 3) Prefer models with high compilation rates. GPT-5 (91.0%), GPT-4o (90.0%), and Qwen (79.0%) produce the most consistently compilable code. CodeLlama’s low compilation rate (34.7%) makes it unsuitable for production smart contract generation. GPT-5 also achieves the lowest vulnerability density (2.22 vulns/100 LOC), making it the strongest choice for security-sensitive contract generation.
- 4) Exercise extreme caution with complex categories. DeFi staking, DEX/AMM, and yield aggregator contracts exhibit 2–3 $\times$ higher vulnerability rates than standard tokens. These categories should receive proportionally more scrutiny.
- 5) Verify security pattern completeness. The phantom guard pattern (Section V) demonstrates that syntactic presence of security constructs does not guarantee functional protection. Manually verify that modifiers and guards contain correct logic.
- 6) Review the second half of long contracts more carefully. Context window degradation causes 4.1 $\times$ higher risky pattern density in the latter portions of complex contracts (>100 LOC).

B. Compilation Quality Analysis

Of 2,400 generated contracts, 834 (34.8%) failed to compile. Root causes are dominated by semantic/type errors (68.5%), followed by import/dependency issues (13.3%), output format contamination such as markdown code fences (7.3%), syntax errors (6.8%), and compiler version incompatibilities (4.1%). LLM-specific patterns are revealing: CodeLlama generates deprecated pre-0.8.x syntax, Gemini suffers from markdown contamination, Claude references bleeding-edge compiler versions, and GPT-4o’s failures are almost purely semantic (80%), indicating the highest Solidity fluency.

C. Implications for LLM Providers and the Security Community

Our findings suggest actionable improvements: (1) Security-focused fine-tuning—incorporate verified, audited smart contract code (e.g., OpenZeppelin) as high-priority training data and downweight unaudited Etherscan code [63]; (2) Post-generation validation—integrate automatic security scanning that flags common patterns (reentrancy, missing access control) before presenting code to users; (3) Security warnings—include explicit disclaimers when generating financial contracts; and (4) Context window management—address the degradation pattern by maintaining security-relevant context throughout long generations.

For the security community, our work motivates LLM-aware security tools that target the phantom guard, inherited vulnerability, and context degradation patterns we identified. Our dataset of 2,400 labeled contracts can serve as a benchmark. The finding that vulnerabilities are intrinsic (not prompt-dependent) motivates research into training data curation for security-critical code generation.

D. Limitations

We acknowledge several limitations: (1) Model evolution—LLM behavior evolves with updates; our results reflect specific versions, though structural patterns (phantom guards, context degradation) likely persist. (2) Prompt representativeness—our 300 prompts across 20 categories may not fully capture real-world prompting patterns. (3) Tool limitations—static analysis produces false positives and negatives; we mitigate through multi-tool analysis and deduplication, with manual validation of 120 contracts underway. (4) Model coverage—eight LLMs are studied; extending to additional models is future work. (5) Language scope—we focus on Solidity; generalization to Rust/Solana, Move/Aptos requires further investigation. (6) Human baseline bias—our 241 Etherscan contracts represent production-grade code but may not capture the full distribution of human-written Solidity; the 54.8% compilation rate introduces selection bias toward simpler contracts. (7) Compilation bias—our compiled-only analysis (65.2%) may miss vulnerability patterns in non-compilable contracts.

E. Threats to Validity

Internal: Prompt design may bias results; we mitigate this with diverse prompts across complexity levels/categories and a consistent system prompt across all models. External: Results may not generalize to multi-turn interactive sessions; we focus on single-shot generation as the most common and highest-risk use case. Construct: Vulnerability counts may not reflect exploitability; we report severity-stratified results and LOC-normalized rates to control for contract size.

Data Availability & Artifact

Our complete reproducibility package is publicly available at: [https://github.com/\[anonymized-for-review\]](https://github.com/[anonymized-for-review]). The artifact includes: (1) all 2,400 LLM-generated contracts with prompt metadata, (2) 241 human baseline contracts (Etherscan-verified, forge-flattened), (3) raw and aggregated analysis results from Slither, Mythril, and Semgrep in structured JSON/CSV, (4) all analysis scripts (dataset generation, vulnerability pipeline, statistical analysis, figure generation), (5) the CWE-mapped vulnerability taxonomy, and (6) instructions for end-to-end reproduction. Model versions, API configurations, and compiler versions are documented to enable exact replication.

Ethical Considerations

Our adversarial prompts are designed to study LLM behavior, not to enable attacks. We follow responsible disclosure practices: findings regarding specific LLM vulnerabilities were reported to respective providers prior to publication. Our dataset contains only synthetically generated contracts; no deployed contracts were tested or exploited. The adversarial prompt categories we document are constructed from patterns already discussed in the security literature [14], [43], and we do not introduce novel attack techniques.

Acknowledgment

[To be added upon acceptance.]

VIII. Conclusion

We presented the first large-scale empirical study of security vulnerabilities in LLM-generated smart contracts. Our analysis of 2,400 contracts across eight LLMs reveals that 64.5% of compilable contracts contain at least one vulnerability, with 26.8% containing high-severity flaws. Compilation rates range from 91.0% (GPT-5) to 34.7% (CodeLlama), and when controlling for contract size, vulnerability density ranges from 2.22 (GPT-5) to 8.43 (Gemini) vulns/100 LOC.

Our most striking finding is the adversarial null result: adversarial prompts do not increase vulnerability rates ($p = 0.997$, Cliff’s $\delta = -0.090$), implying that vulnerabilities are intrinsic to LLM code generation rather than prompt-dependent. We also identified three novel LLM-specific vulnerability patterns—phantom guards, inherited

vulnerabilities, and context window degradation—that require LLM-aware detection tools.

Comparison with 241 verified Ethereum mainnet contracts reveals that LLM-generated vulnerability density is comparable to human-written code (2.69 vulns/100 LOC), with nearly identical high-severity proportions (16.0% vs. 15.7%). The primary risk is not inferior code quality but the security infrastructure developers bypass when trusting LLM output.

We release our full dataset, analysis pipeline, and vulnerability taxonomy. Future work includes extending to additional LLMs and languages (Rust/Solana, Move/Aptos), runtime analysis through fuzzing, multi-turn generation sessions, and LLM-aware security tool development.

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TABLE X: Vulnerability Rates by Contract Category (Top 10, Compiled Only)

Category	Mean Vulns	% High	Most Common CWE
DeFi Staking	6.77	46.2%	CWE-841
Timelock	6.31	25.9%	CWE-829
Auction	6.05	24.1%	CWE-829
Crowdfunding/ICO	5.51	17.6%	CWE-829
Yield Aggregator	4.94	35.2%	CWE-841
DEX/AMM	4.73	37.8%	CWE-841
DeFi Lending	4.20	25.3%	CWE-841
Governance/DAO	3.49	11.4%	CWE-829
ERC721 NFT	3.36	64.4%	CWE-841
Proxy/Upgradeable	3.30	50.0%	CWE-829

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Appendix A

Contract Category Analysis

Vulnerability rates vary substantially across contract categories (Kruskal-Wallis $H = 466.5$, $p < 10^{-87}$). Table X presents the per-category analysis for the top 10 categories ranked by mean vulnerability count.

A. DeFi Contracts: Highest Vulnerability Rates

DeFi staking contracts exhibit the highest mean vulnerability rate (6.77 per contract), followed by timelock contracts (6.31) and auctions (6.05). These categories involve complex state management, token transfer logic, and reward calculation—areas where LLMs consistently introduce reentrancy and timestamp dependence vulnerabilities. DeFi staking also has the highest high-severity rate among DeFi categories (46.2%).

DEX/AMM contracts have both high vulnerability rates (4.73) and a substantial proportion of high-severity vulnerabilities (37.8%), making them a particularly dangerous category for direct financial loss. LLMs frequently generate AMM contracts with unprotected swap functions and missing slippage checks.

B. ERC721, Flash Loan, and Proxy Contracts: High-Severity Concentration

While ERC721 NFT contracts have a moderate mean vulnerability count (3.36), they exhibit the second-highest high-severity rate (64.4%) among all categories. This is driven by access control vulnerabilities in minting and transfer functions. Flash loan contracts (3.06 mean vulns) show the highest high-severity concentration (69.2%), reflecting the critical nature of callback validation and reentrancy prevention in flash loan architectures [23]. Proxy/upgradeable contracts show a 50.0% high-severity rate, primarily from unprotected initializer functions and missing upgrade authorization checks.

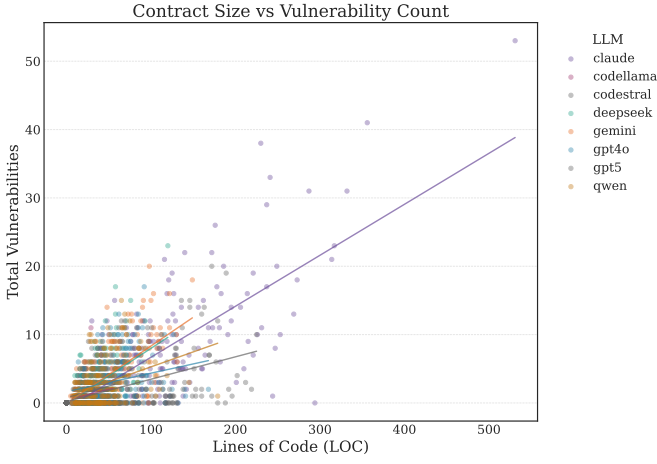


Fig. 9: Contract size (LOC) vs. vulnerability count with per-LLM regression lines. Larger contracts contain more vulnerabilities, but the slope varies by model—Claude generates the largest contracts with correspondingly more findings.

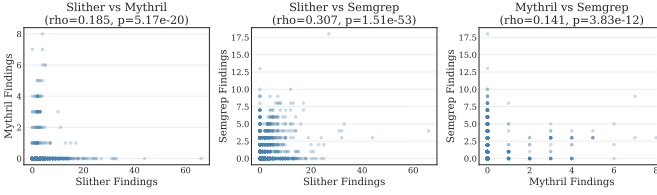


Fig. 10: Pairwise tool agreement scatter plots. Low Spearman correlations across all three tool pairs confirm complementary detection capabilities.

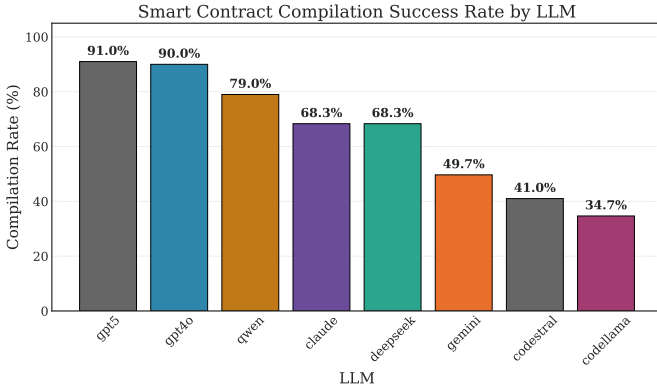


Fig. 7: Compilation success rate by LLM. GPT-5 and GPT-4o achieve >90%, while code-specialized models CodeLlama and Codestral compile fewer than half their outputs.

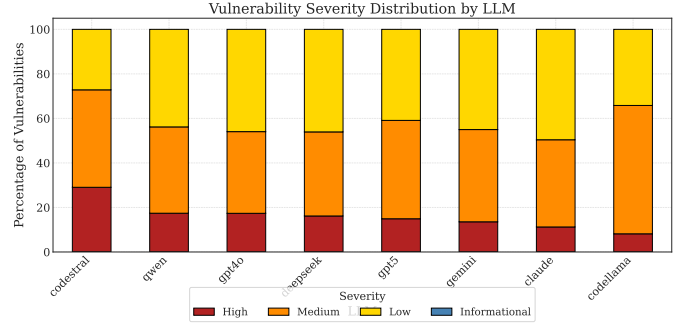


Fig. 8: Vulnerability severity distribution by LLM. Codestral has the highest proportion of high-severity findings (29.5%), while CodeLlama has the lowest (8.6%). Low-severity findings dominate across all models.

C. Standard Contracts: Lower but Non-Zero Risk

The relatively lower vulnerability rates in ERC20 tokens (0.41) and ERC1155 multi-tokens (0.65) suggest that LLMs have been trained on sufficient examples of these well-established patterns. However, even these “simple” contracts exhibit high-severity vulnerabilities in 3.8% and 9.1% of cases respectively, indicating that no contract category should be considered safe by default.

Access control contracts have the lowest mean vulnerability rate (0.74), likely because their focused scope limits the attack surface. Cross-chain messaging contracts are similarly low (1.52) with a very low high-severity rate (3.2%), suggesting that LLMs handle message-passing patterns reasonably well when the contracts are simple enough to compile.

Appendix B Supplementary Figures

